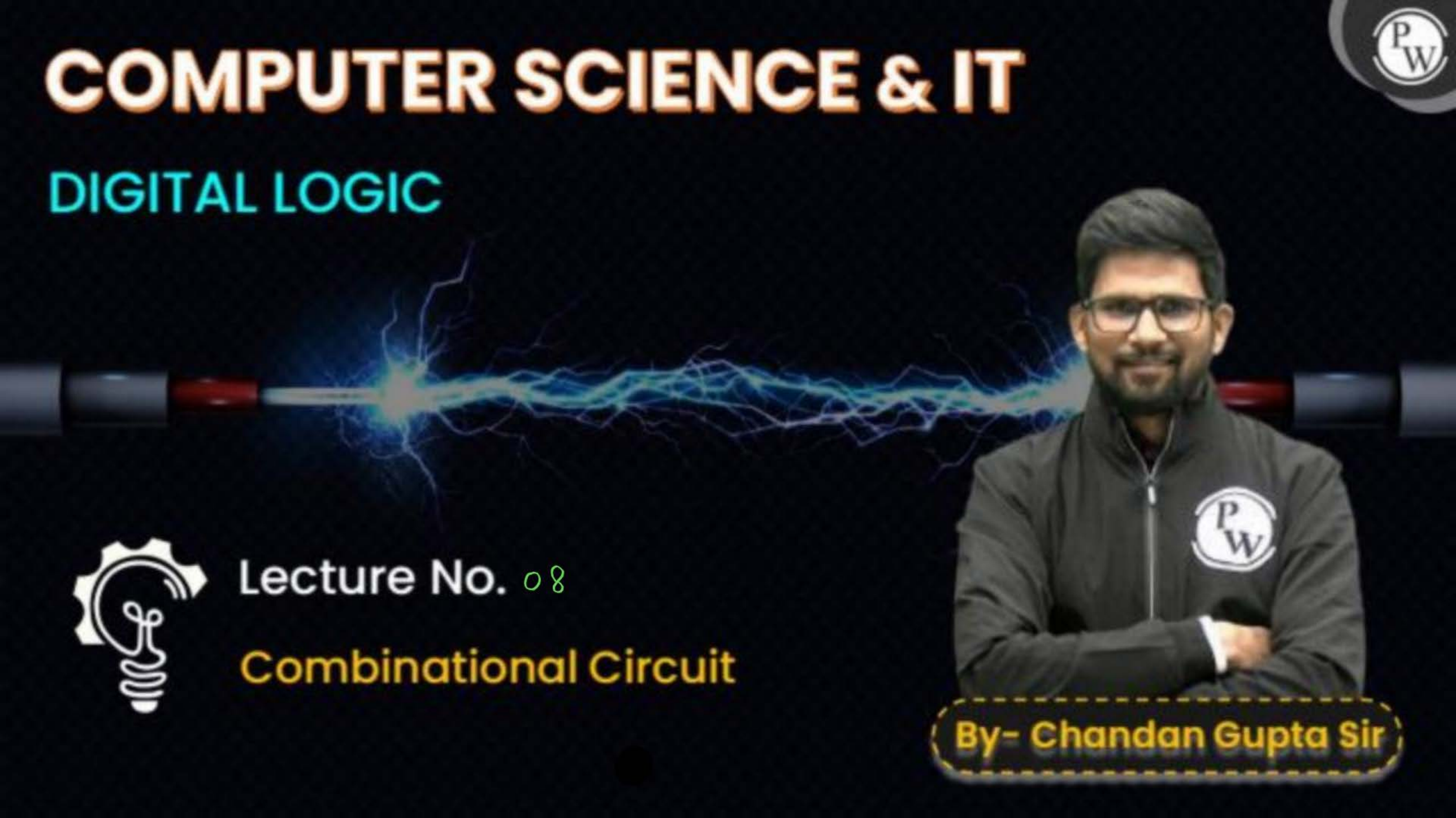
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 08

Combinational Circuit

By- Chandan Gupta Sir



K-Map



- Comparator circuit

Question

If the function W, X, Y, Z are as :

$$W = R + \bar{P}Q + \bar{R}S$$

$$X = PQ\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S} + \bar{P}\bar{Q}\bar{R}\bar{S}$$

$$Y = RS + \overline{PR + P\bar{Q} + \bar{P}\bar{Q}}$$

$$Z = R + S + \overline{PQ + \bar{P}\bar{Q}\bar{R} + P\bar{Q}\bar{S}}$$

Then which of the following is/are correct ?

(a) $W = \bar{X}$

(b) $W = \bar{Z}$

(c) $\bar{W} = \bar{Z}$

(d) $\bar{X} = Z$

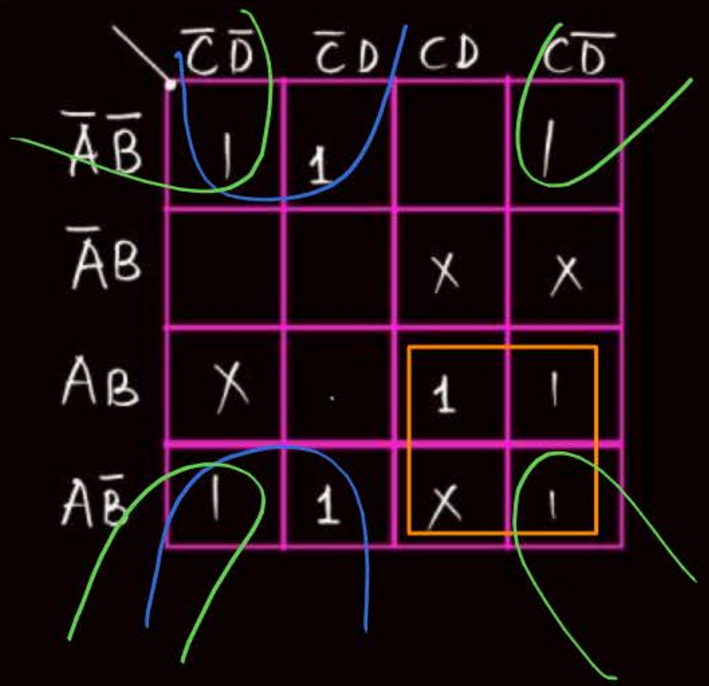
Ans:

$W = \bar{X}$ or $\bar{W} = X$

$W = Z$

& $W = Z = \bar{X}$

Q.



$\bar{B}\bar{C} + \bar{B}\bar{D} + AC$

or

$\bar{B}\bar{C} + \bar{B}\bar{D} + BC$

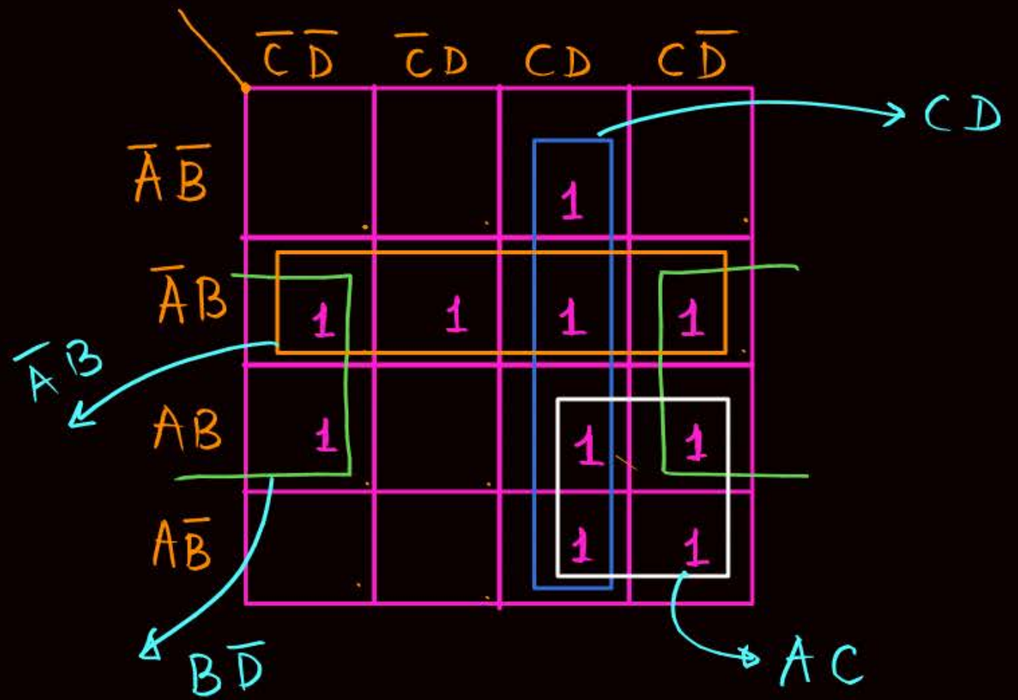
or $\bar{B}\bar{C} + C\bar{D} + AC$

or

$\bar{B}\bar{C} + C\bar{D} + BC$

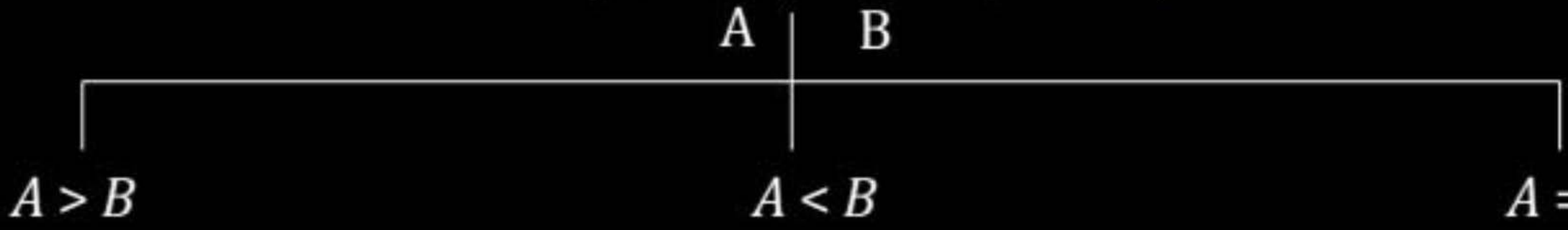
#Q. $f(A,B,C,D) = \overline{\bar{A}\bar{D} + \bar{B}\bar{C} + AB\bar{C}} + B\bar{D}$

Minimize this boolean function using K-Map.



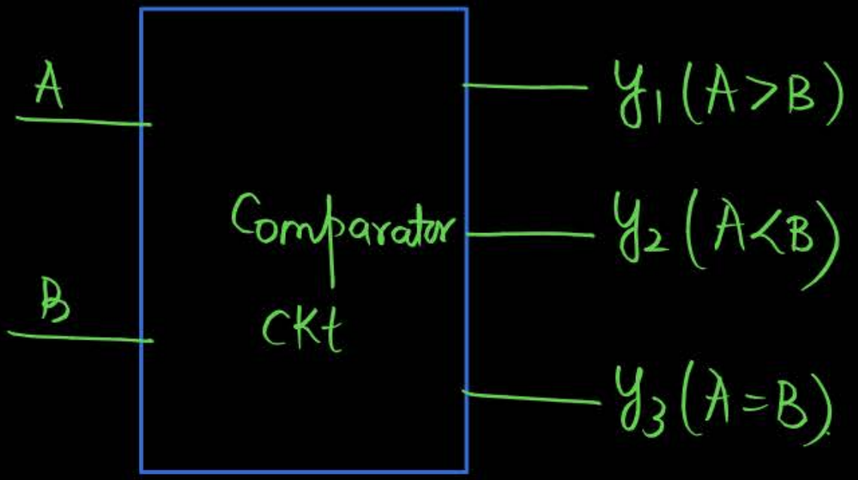
$= \bar{A}B + B\bar{D} + CD + AC$

Comparator [9mp.]



- A and B are single bit number then how to design comparator ?

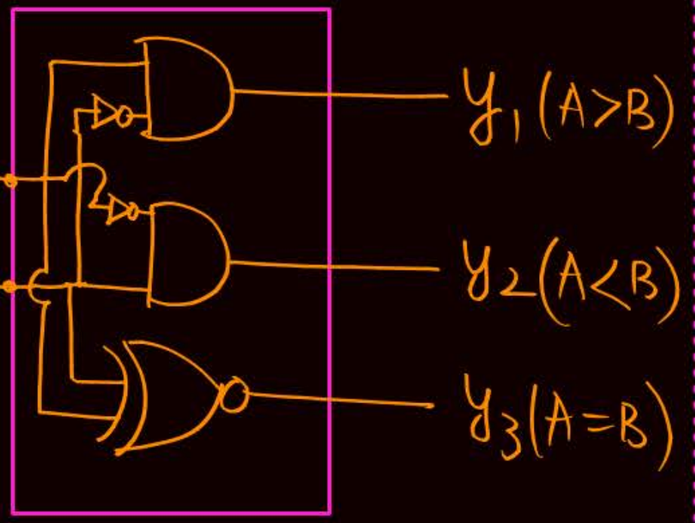
A	B	$y_1 (A>B)$	$y_2 (A<B)$	$y_3 (A=B)$
0	0	0	0	1
0	1	0	1	0
1	0	1	0	0
1	1	0	0	1

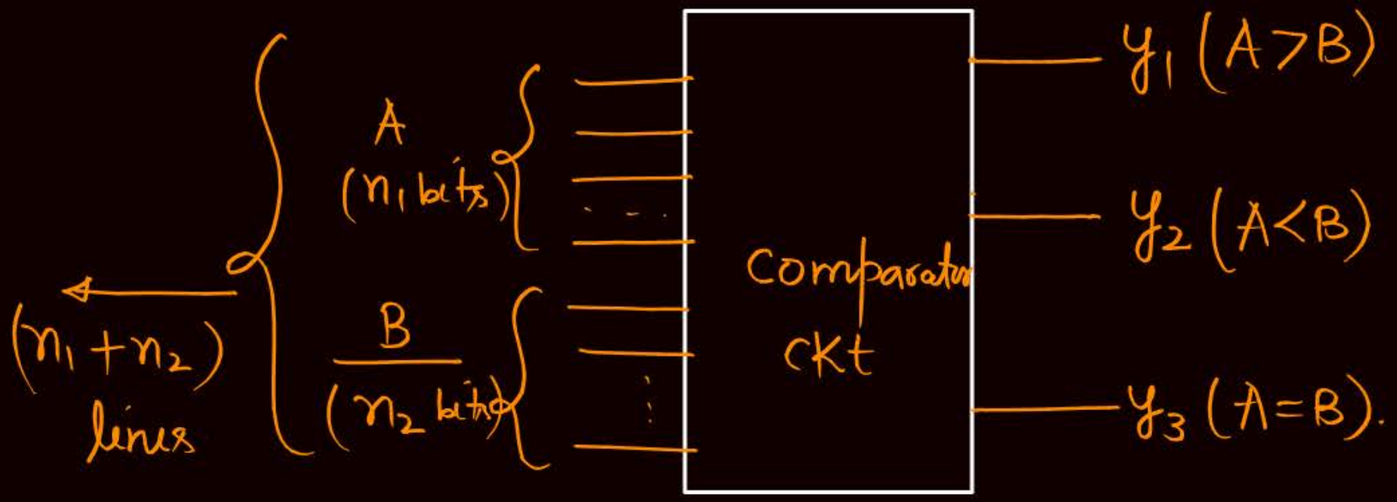


$$y_1[A>B]\,[A,B] = \Sigma(2) = A\cdot\overline{B}$$

$$y_2[A<B]\,[A,B] = \Sigma(1) = \overline{A}\cdot B$$

$$y_3[A=B]\,[A,B] = \Sigma(0,3)$$
$$= \overline{A}\cdot\overline{B} + A\cdot B$$
$$= A \odot B.$$





$A = 3\text{ bit} = a_2 a_1 a_0$

$B = 3\text{ bit} = b_2 b_1 b_0$

$$y_1(A>B) = (a_2 > b_2) \text{ or } (a_2 = b_2).(a_1 > b_1) \text{ or } (a_2 = b_2).(a_1 = b_1).(a_0 > b_0)$$

$$= a_2 \cdot \overline{b_2} + (a_2 \odot b_2) \cdot (a_1 \overline{b_1}) + (a_2 \odot b_2) \cdot (a_1 \odot b_1)(a_0 \overline{b_0})$$

$A = 101$

$B = 011,\ 100$

$$y_2(A<B) = (a_2 < b_2) \text{ or } (a_2 = b_2)(a_1 < b_1) \text{ or } (a_2 = b_2)(a_1 = b_1)(a_0 < b_0)$$

$$= \overline{a_2} b_2 + (a_2 \odot b_2)(\overline{a_1} \cdot b_1) + (a_2 \odot b_2)(a_1 \odot b_1)\overline{a_0} \cdot b_0$$

$$y_3(A=B) = (a_2 = b_2) \cdot (a_1 = b_1) \cdot (a_0 = b_0)$$

$$= (a_2 \odot b_2)(a_1 \odot b_1) \cdot (a_0 \odot b_0)$$

H.W.

$A = a_3\, a_2\, a_1\, a_0$

$B = b_3\, b_2\, b_1\, b_0$

$y_1 (A > B) =$

$y_2 (A < B) =$

$y_3 (A = B) =$

H.W.

$A = a_2\, a_1\, a_0$

$B = b_1\, b_0$ →

$y_1 (A > B) =$

$y_2 (A < B) =$

$y_3 (A = B) =$

A	B
2 bit	2 bit
0	0
1	1
2	2
3	3

Total bits = 4 bits $\longrightarrow$ total combination $= 2^4 = 16$

$N_1 (A > B) = 0 + 1 + 2 + 3 = 6$

$N_2 (A < B) = 0 + 1 + 2 + 3 = 6$

$N_3 (A = B) = 4 = 2^2$

$N_1 + N_2 + N_3 = 2^4 = 16$

A	B
n bit	n bit

total bits $\Rightarrow$ 2n bits

Total comb. $= 2^{2n}$

0, 1, 2, ... $(2^n - 1)$ 0, 1, 2, ... $(2^n - 1)$

$N_1 (A > B) = N_2 (A < B) = 0 + 1 + 2 + 3 + \cdots (2^n - 1)$

$N_3 (A = B) = 2^n$

A	B
2 bit	3 bit
0	0
1	1
2	2
3	3
	4
	5
	6
	7

Total bits = 5 bits $\longrightarrow$ total comb. $= 2^5 = 32$

$N_1 (A > B) = 0 + 1 + 2 + 3 = 6$

$N_2 (A < B) = 4 + 5 + 6 + 7 = 22$

$N_3 (A = B) = 4$

$N_1 + N_2 + N_3 = 32 = 2^5$

Case-I

A → n_1 bits
B → n_2 bits
⟶ total (n_1+n_2) bits
$2^{n_1+n_2}$ comb.

$(n_1 < n_2)$

A	B
0	0
1	1
⋮	⋮
$(2^{n_1}-1)$	⋮
	$(2^{n_2}-1)$

$N_1(A>B) = 0+1+2+\cdots (2^{n_1}-1)$

$N_3(A=B) = 2^{n_1}$

$N_2(A<B) = 2^{(n_1+n_2)} - N_1 - N_3$

Case-II $n_1 > n_2$

A = n_1 bits
B = n_2 bit
⟶ Total bits = n_1+n_2
Total comb. = $2^{n_1+n_2}$

$N_2(B>A) = N_2(A<B)$

$= 0+1+2+3\cdots (2^{n_2}-1)$

$N_3(A=B) = 2^{n_2}$

$N_1 = 2^{n_1+n_2} - N_2 - N_3$
$(A>B)$

6. $A = 4$ bit
   $B = 7$ bit
   $\Longrightarrow$ total bits $= 11$
   Total Comb. $= 2^{11} = 2048$

$$N_1(A>B) = 0+1+2+\cdots+15 = \frac{15(16)}{2} = 120$$

$$N_3(A=B) = 2^4 = 16$$

$$N_2(A<B) = 2048 - N_1 - N_3 = 2048 - 120 - 16 = 1912$$

- A = 8 bits
  B = 5 bits
  ——— total bits = 13
  total comb. $= 2^{13} = 8 \times 1024 = 8192$

$N_1 (A > B)$ ——— 7664.

$N_2 (A < B) = 0 + 1 + 2 + \cdots 31 = \frac{31 \times 32}{2} = 31 \times 16 = 496$

$N_3 (A = B) = 2^5 = 32$

$N_1 (A > B) = 8192 - 496 - 32 = 7664$

- How many terms will be there in POS expression of
  $Y_1 (A > B)$ 528.
  $Y_2 (A < B)$ 7696
  $Y_3 (A = B)$ 8160

Question

We have two number A and B both A and B are 2-bit numbers then in how many combination A > B.

(a) 5

(b) 6

(c) 7

(d) 8

H.W.

Question

We have two 4-bit number A and B then number of combinations in which A < B _____

Number of combinations in which A = B _____.

H.W.

H.W.

Q.1. A = 6 bits

B = 2 bits

$N_1 (A > B)$ ——————

$N_3 (A = B)$ ——————

Q.2. A = 8 bits

B = 12 bits

No. of min-term (SOP-term) in logical O/P of $W (A = B)$ ——.

$Z (A < B)$ ——.



2 Minute Summary

- Comparator ckt.

Thank you

GW

Soldiers !

